

MAYANK PANDEY

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PROFESSIONAL SUMMARY

Innovative and results-driven Full-Stack & AI Developer with expertise in Machine Learning, Computer Vision, and Full-Stack development. Experienced in building production-ready AI applications, integrating deep learning models, and developing scalable distributed web platforms using React 19, Next.js, Node.js, Spring Boot, and FastAPI microservices.

TECHNICAL SKILLS

AI/ML	Scikit-learn, XGBoost, TensorFlow, Keras, OpenCV, MTCNN, Transfer Learning (VGG16), NLP (TF-IDF), LLMs (Gemini, Ollama)
Web Development	React, Next.js, Vite, Node.js, Express, FastAPI, Spring Boot, Flask, Django
Programming	Python, C++, Java, SQL, JavaScript (ES6+), TypeScript
CS Fundamentals	Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Tools	Git, Docker, PostgreSQL, MongoDB, MySQL, Vercel, Render

PROJECTS

SentinelAI - Real-time Fraud Detection Platform

[GitHub](#) | [Demo](#)

TypeScript, Next.js, FastAPI, Spring Boot, XGBoost, PostgreSQL, Docker

- Developed a high-performance fraud detection platform integrating rule-based heuristics with an XGBoost/Random Forest ML pipeline for real-time transaction risk scoring and anomaly detection.
- Architected a dual-backend system: Spring Boot for core business logic and transaction handling, with a dedicated FastAPI microservice serving sub-50ms ML inference over REST.
- Built a live monitoring dashboard in Next.js and TypeScript, enabling administrators to track flagged transactions, analytics, and anomaly trends instantly.
- Deployed the full-stack containerized platform to Render with CI-friendly testing, exposing a live working demo.

careerpilot-ai - Automated Career Platform

[GitHub](#) | [Demo](#)

React 19, Node.js, Express, Vite, Local AI (Ollama + Qwen3), Tailwind CSS

- Engineered a comprehensive career preparation platform featuring ATS resume analysis, keyword-gap detection, and intelligent job matching algorithms.
- Implemented a mock interview simulator powered entirely by locally-hosted AI models (Ollama + Qwen3), ensuring 100% free and private AI inference with zero API cost.
- Designed the resume-parsing and feedback pipeline with regex heuristics and TF-IDF to generate section-wise actionable improvement suggestions.
- Optimized frontend performance utilizing React 19 and Vite for a blazing-fast, low-latency user experience, deployed on Vercel.

Smart Expense Tracker

[GitHub](#) | [Demo](#)

React 18, Express, PostgreSQL, Gemini AI, Chart.js

- Built a full-stack expense tracking solution featuring automated receipt OCR powered by the Gemini AI API for structured line-item extraction.
- Implemented comprehensive financial tools including monthly budget alerts, category-wise spend analytics, and CSV bank statement parsing for seamless data import.
- Designed a normalized PostgreSQL schema for transactions, budgets, and categories to support fast querying, aggregation, and reporting.
- Achieved 100% integration test coverage ensuring enterprise-level reliability, successfully deployed across Vercel and Render.

Deepfake Detection Model Comparison

[GitHub](#)

Python, TensorFlow/Keras, OpenCV, MTCNN, Computer Vision

- Benchmarked 6 deep learning architectures (VGG16+MTCNN, CNN, LSTM, CNN+LSTM, InceptionV3, InceptionV3+Attention) for binary deepfake image classification across 2 datasets (12K+ images).
- Identified CNN+InceptionV3+Attention as the top performer, achieving 99.66% training and 100% validation accuracy on Dataset 1.
- Evaluated all models on accuracy and loss across train/validation splits, visualizing comparative performance trends to surface overfitting and generalization gaps.
- Structured the codebase into reproducible notebooks with a shared MTCNN preprocessing pipeline for fair cross-model comparison.

EDUCATION

VIT Bhopal University

2023 – 2027

B.Tech CSE with Specialisation in AIML — CGPA: 8.44/10

Bhopal, India

CERTIFICATIONS

NPTEL — Introduction to Internet of Things

NPTEL (IIT Kharagpur) — Apr 2026

Google IT Support Certificate

Google Career Certificates — Jan 2026

AWS Certification Course

Intellipaat — Sep 2025

Applied Machine Learning in Python

University of Michigan — Nov 2025

NPTEL — Cloud Computing

NPTEL (IIT Kharagpur) — Apr 2025